

FLATRUET, Sweden

WMO: 023030

Lat: 62.7291N

Lon: 12.7264E

Elev: 932

StdP: 90.62

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.5	-18.4	-24.2	0.5	-17.5	-22.0	0.6	-15.6	21.0	-3.2	19.3	-3.7	2.0	0	0.867

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	20.2	12.6	18.4	11.8	16.5	11.1	14.1	17.6	13.2	16.3	12.2	15.1	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	10.2	14.8	11.7	9.6	14.0	10.8	9.0	13.3	42.5	17.8	39.9	16.2	37.2	15.2	17.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.5	11.5	DB	-22.9	22.8	3.1	1.4	-25.2	23.8	-27.0	24.6	-28.7	25.4	-31.0	26.5	(4)
(5)			WB	-23.3	15.2	2.6	1.5	-25.2	16.3	-26.8	17.2	-28.2	18.1	-30.2	19.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.3	-8.1	-8.0	-5.8	-1.4	2.9	7.9	10.7	10.3	5.5	0.4	-4.0	-7.1	(6)	
	DBStd	7.90	4.67	4.79	4.41	3.75	3.82	3.87	3.04	3.25	3.35	3.97	4.41	4.79	(7)	
	HDD10.0	3654	560	505	491	341	222	86	27	37	138	299	419	529	(8)	
	HDD18.3	6569	818	738	749	591	477	313	236	249	384	557	669	788	(9)	
	CDD10.0	127	0	0	0	0	3	24	49	46	4	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	2	0	0	0	0	0	1	1	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	5.1	5.4	5.4	6.1	5.5	5.2	5.1	4.6	4.2	5.0	5.1	4.9	4.7	(14)	
(15) Precipitation	PrecAvg	624	48	35	32	32	48	70	87	82	52	52	48	38	(15)	
	PrecMax	785	101	59	47	78	95	132	168	136	91	118	96	58	(16)	
	PrecMin	509	8	6	10	8	16	24	25	38	17	21	17	15	(17)	
	PrecStd	77	23	14	11	16	22	30	35	28	23	25	21	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	2.7	4.3	9.2	18.4	21.8	22.4	21.2	16.5	10.4	6.7	2.9	(19)	
		MCWB	-0.3	-0.8	0.7	4.9	11.4	12.4	14.9	12.9	9.9	7.5	3.4	1.3	(20)	
	2%	DB	0.4	0.5	2.4	6.7	14.4	19.9	20.0	19.2	14.3	8.3	5.0	1.8	(21)	
		MCWB	-0.7	-0.8	0.6	2.5	8.3	11.5	13.5	11.8	9.1	6.0	3.1	0.6	(22)	
	5%	DB	-0.7	-0.6	1.2	5.2	11.0	17.3	18.1	17.5	12.1	6.9	3.0	0.6	(23)	
		MCWB	-1.4	-1.6	-0.4	1.9	6.7	9.9	12.8	11.7	8.8	5.0	1.9	-0.5	(24)	
	10%	DB	-1.9	-1.9	0.0	3.7	8.6	14.6	16.1	15.6	10.5	5.6	1.6	-0.8	(25)	
		MCWB	-2.6	-2.8	-1.4	1.2	5.0	9.2	11.6	11.4	8.0	4.2	0.8	-1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	0.8	1.9	5.5	12.0	13.9	15.5	15.4	11.6	8.6	5.1	2.0	(27)	
		MCDB	0.9	1.4	2.7	8.2	17.0	17.6	20.5	18.5	14.0	9.6	5.9	2.4	(28)	
	2%	WB	-0.4	-0.6	1.1	3.6	9.4	12.7	14.2	14.0	10.3	6.6	3.5	0.7	(29)	
		MCDB	0.1	-0.1	2.1	5.8	13.6	17.5	18.4	16.8	12.6	7.7	4.4	1.5	(30)	
	5%	WB	-1.3	-1.5	-0.1	2.5	7.3	11.4	13.4	12.9	9.4	5.4	2.1	-0.3	(31)	
		MCDB	-0.8	-0.8	1.0	4.3	10.1	15.7	16.8	15.4	11.5	6.5	2.7	0.4	(32)	
	10%	WB	-2.6	-2.8	-1.3	1.6	5.7	9.9	12.5	11.9	8.2	4.4	0.8	-1.6	(33)	
		MCDB	-1.9	-1.8	-0.3	3.2	8.2	14.1	15.3	14.1	10.0	5.5	1.4	-0.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	4.8	5.1	5.7	6.0	7.4	7.0	6.7	5.8	4.5	4.0	4.7	(35)	
		MCDBR	4.5	5.0	5.3	7.0	9.5	11.0	9.7	9.1	8.3	6.2	5.0	5.6	(36)	
	5% WB	MCWBR	4.4	4.7	4.3	4.3	5.3	5.3	4.8	4.3	4.7	4.4	4.3	5.4	(37)	
		MCDWR	4.5	4.9	5.2	6.4	8.3	9.7	9.0	7.6	7.4	5.8	4.9	5.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.187	0.243	0.271	0.302	0.313	0.311	0.323	0.312	0.286	0.269	0.198	0.166	(40)	
		taud	2.053	2.255	2.269	2.319	2.473	2.529	2.502	2.519	2.580	2.482	2.117	1.964	(41)	
	Ebn at Noon		509	735	844	888	908	916	892	874	839	710	481	313	(42)	
		Edh at Noon	30	58	84	100	94	91	91	82	63	47	28	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.21	0.80	2.25	4.10	4.72	5.02	4.66	3.58	2.24	0.97	0.29	0.11	(44)	
	RadStd		0.02	0.12	0.20	0.34	0.34	0.44	0.55	0.33	0.24	0.13	0.05	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-142	-178	N/A	N/A

Nomenclature: See separate page